

Circuits — Full Practice

Physics Standard 7.1 · Ohm's Law and Circuits

Part 1 builds the foundation: one resistor, then pure series, then pure parallel. Part 2 is the hard set, ending at the exact Level 4 target of complex mixed circuits with multiple parallel branches.

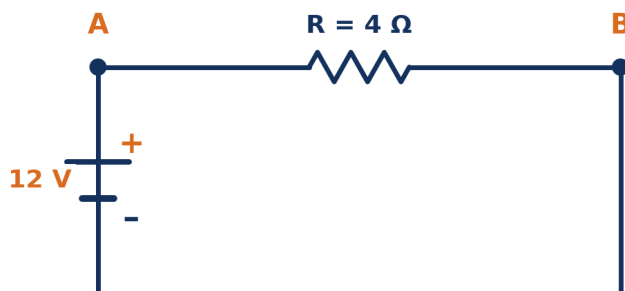
Every circuit here collapses using series and parallel rules alone. No junction rule. No loop rule. Two components are in **parallel** only if both ends touch the same pair of nodes. Two are in **series** only if they share a node with nothing else attached. Points joined by bare wire are one node.

Show every step and carry units throughout.

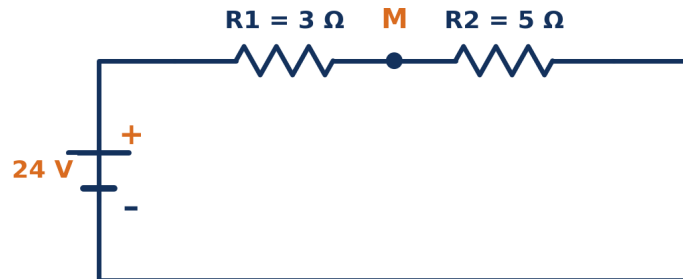
Part 1 · Foundations

Level 1 and 2. One resistor, then series, then parallel. No tricks, no hidden structure.

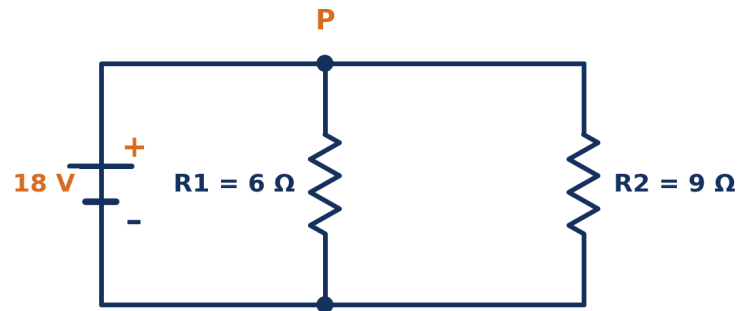
1. A single resistor is connected across a battery.
 - (a) Find the current in the circuit.
 - (b) State the voltage across the resistor and explain in one line why it equals the battery voltage.
 - (c) How much charge passes through the resistor in 5.0 seconds?



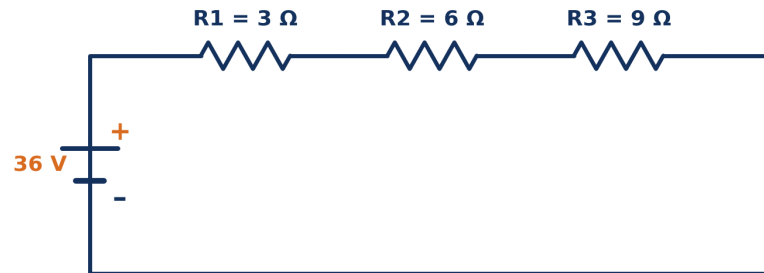
2. Two resistors in series.
- (a) Find the equivalent resistance.
 - (b) Find the current in the circuit.
 - (c) Find the voltage across each resistor and check that they add to 24 V.



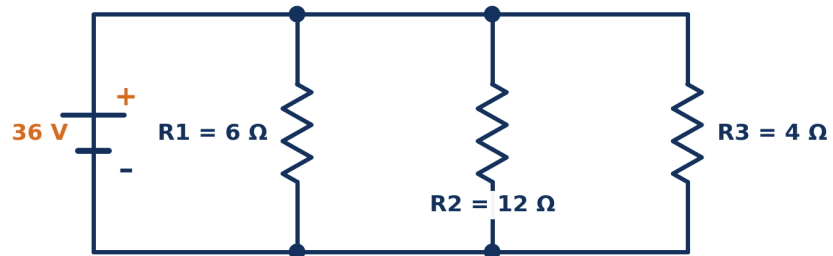
3. Two resistors in parallel.
- (a) Find the equivalent resistance.
 - (b) State the voltage across each resistor.
 - (c) Find the current through each resistor and the total current from the battery.



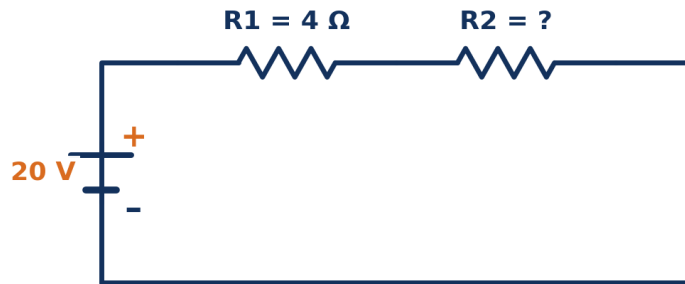
4. Three resistors in series.
- (a) Find the equivalent resistance and the current.
 - (b) Find the voltage across each resistor.
 - (c) Which resistor takes the largest share of the voltage, and why?



5. Three resistors in parallel.
- (a) Find the equivalent resistance.
 - (b) Find the current through each resistor and the total.
 - (c) Which resistor carries the largest current, and why?



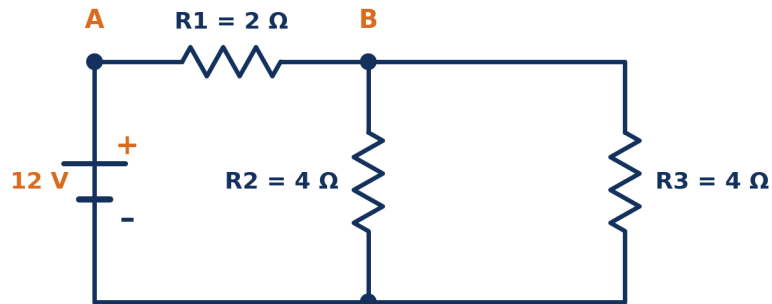
6. Two resistors in series. An ammeter placed anywhere in the loop reads 2.0 A.
- (a) Find the equivalent resistance of the circuit.
 - (b) Find R_2 .
 - (c) Find the voltage across each resistor.



ammeter in the loop reads 2.0 A

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7. No diagram.
- (a) Two resistors in parallel combine to $4.0\ \Omega$. One of them is $12\ \Omega$. Find the other.
 - (b) Three identical $30\ \Omega$ resistors are placed in parallel. Find the equivalent resistance.
 - (c) State in one sentence which quantity every component shares in a series circuit, and which quantity every component shares in a parallel circuit.
 - (d) Explain in one sentence why a parallel combination is always smaller than its smallest branch.
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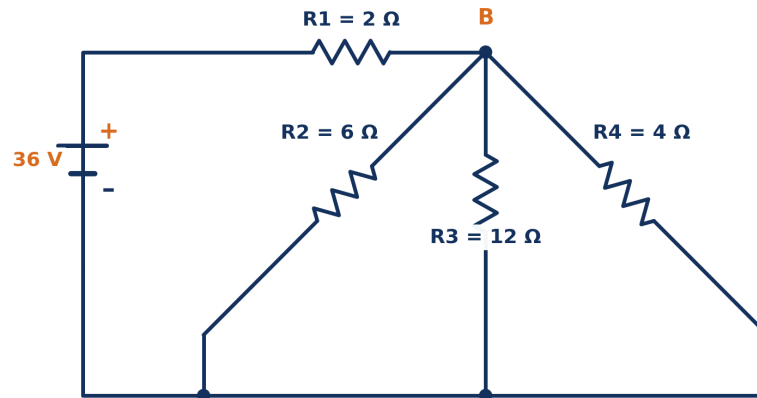
8. One series resistor feeding a parallel pair. This is the simplest mixed circuit there is.
- (a) Find the equivalent resistance and the total current.
 - (b) Find the voltage across R_1 and the voltage across the parallel pair.
 - (c) Find the current through R_2 and through R_3 , and check them against the total.



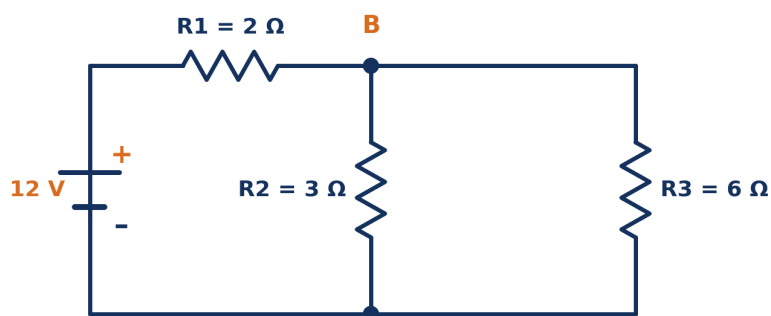
Part 2 • Mixed Circuits and Hidden Structure

Level 3 and 4. Nested networks, shorted branches, dead branches, and circuits whose drawing hides their structure. Redraw by node before writing an equation.

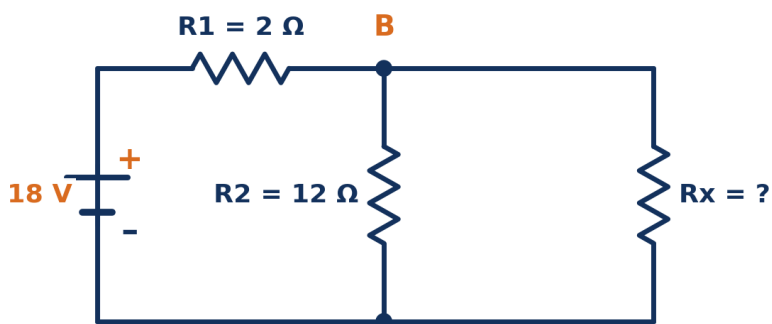
9. Find the equivalent resistance and complete an IRV chart for all four resistors. Verify your branch currents add to the total.



10. For the circuit below, calculate the power dissipated by every resistor and rank them largest to smallest. State which quantity, current or voltage, each resistor shares with its neighbour.

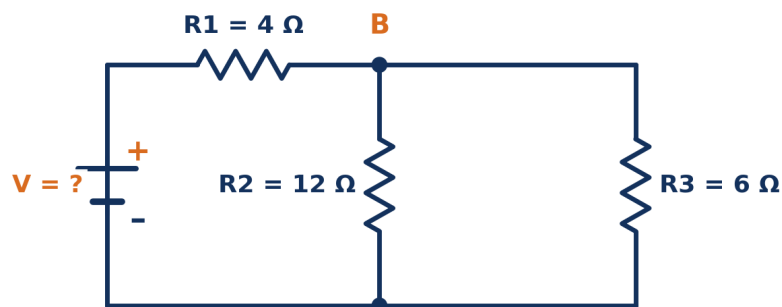


- 11.** An ammeter placed at the battery reads 3.0 A. Find R_x , then find the current through R_2 and through R_x separately.



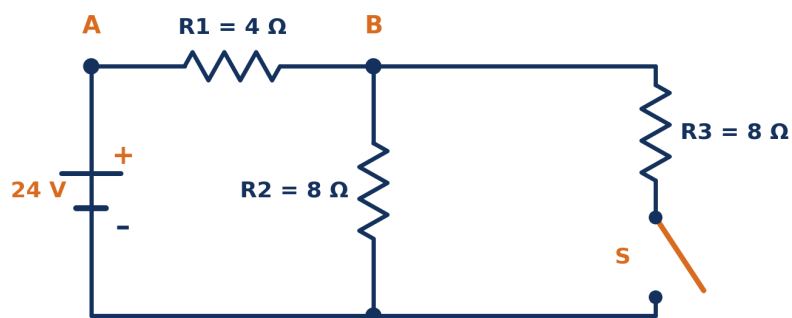
ammeter at the battery reads 3.0 A

12. An ammeter placed in the R_3 branch reads 1.0 A. Find the source voltage V , the total current, and the voltage across R_1 .

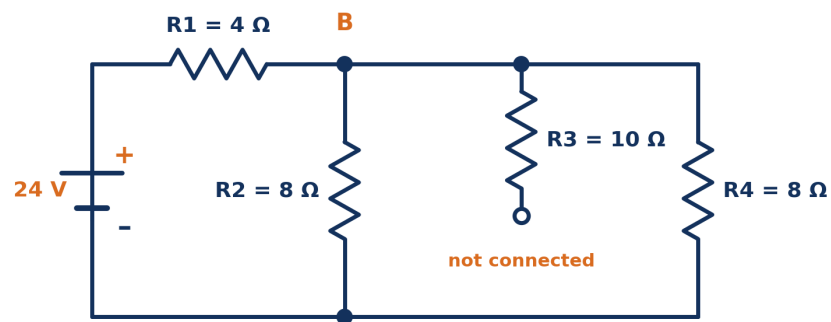


ammeter in the R_3 branch reads 1.0 A

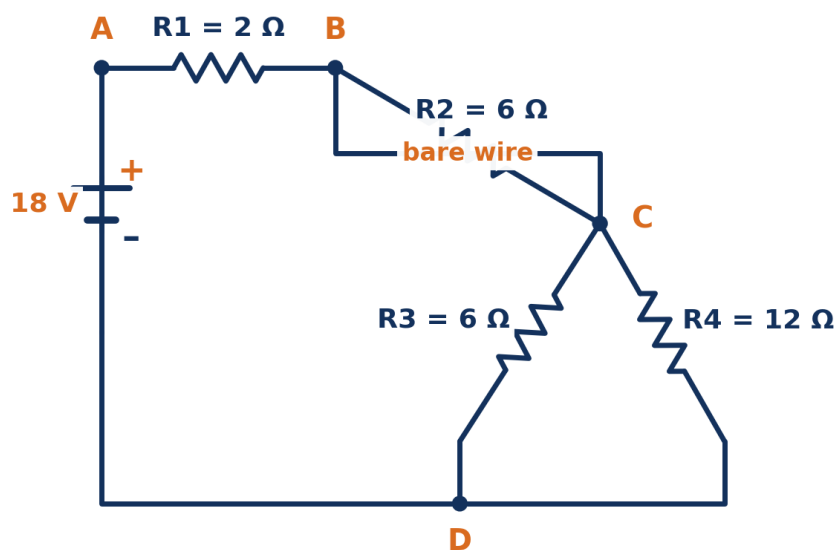
- 13.** Switch S is initially open.
- (a) Find the total current and the voltage at node B.
 - (b) Close S. Find the new total current and the new voltage at node B.
 - (c) Explain why closing a switch that adds a resistor *increases* the current the battery delivers.



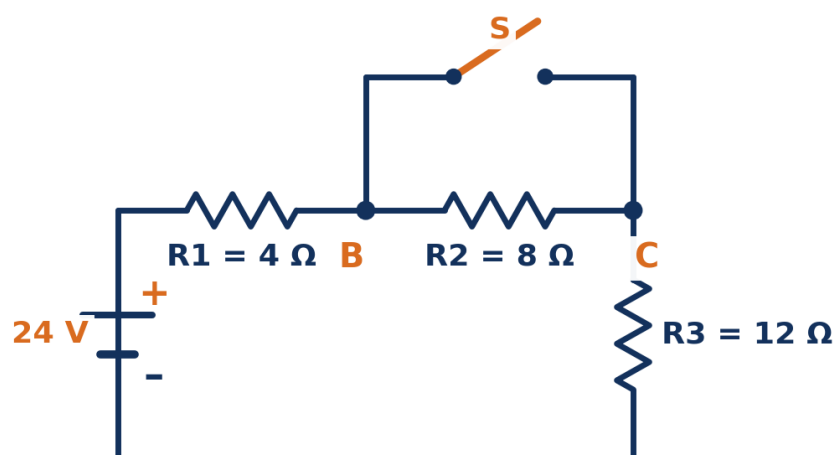
14. R_3 has one end soldered to the top rail and the other left hanging in open air. Find the equivalent resistance, the total current, the current through R_3 , and the voltage across R_3 .



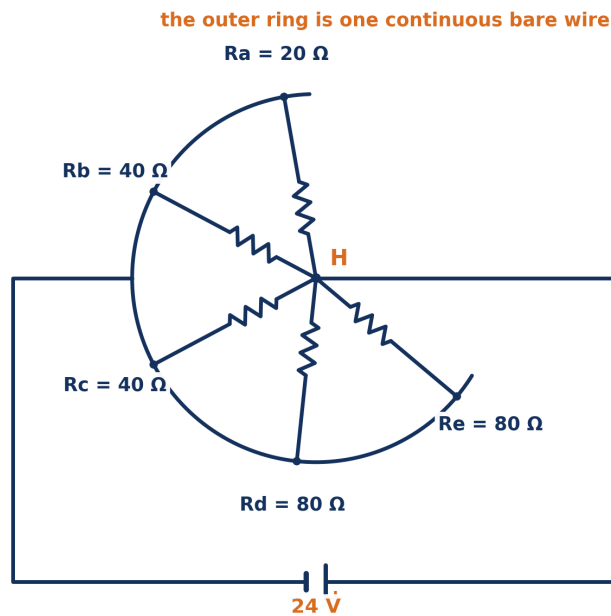
- 15.** A bare wire runs from node B to node C, bypassing R_2 entirely.
- (a) Find the current through R_2 and justify it in one sentence.
 - (b) Find the equivalent resistance and the total current.
 - (c) Find the current through R_3 and R_4 .
 - (d) The bare wire is now cut. Find the new equivalent resistance.



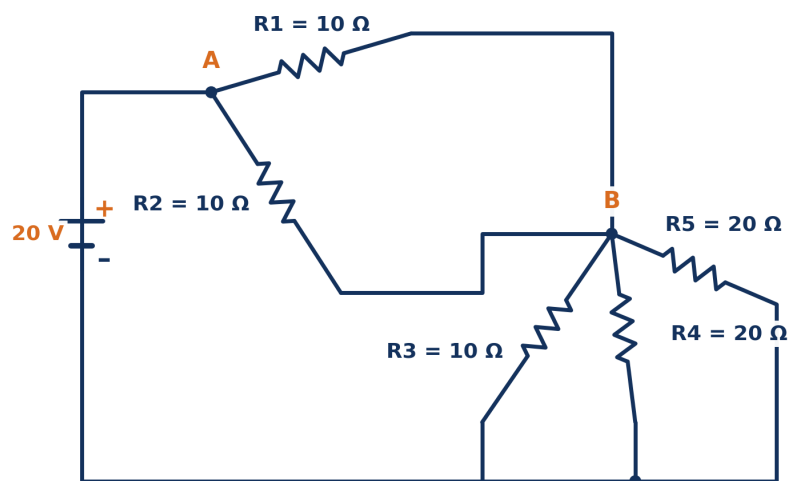
- 16.** Switch S connects node B directly to node C.
- (a) With S open, find the total current and the voltage across R_2 .
 - (b) With S closed, find the total current and the voltage across R_2 .
 - (c) By what factor does the power dissipated in R_3 change?



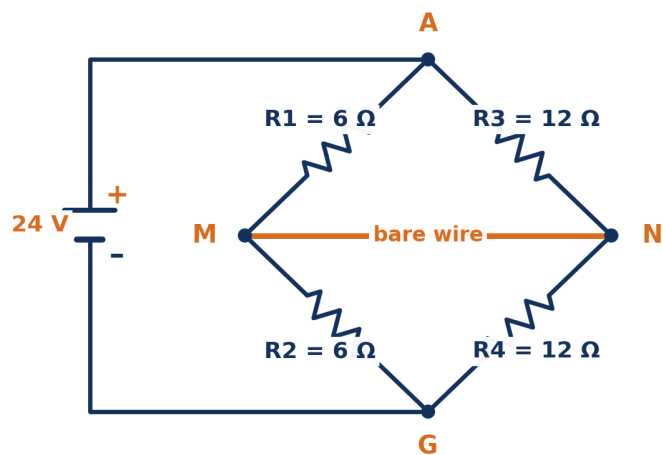
17. Five resistors radiate from hub H to a continuous bare outer ring. Redraw this in a standard rectangular form, then find the equivalent resistance, the total current, and the current in each of the five resistors.



18. The resistors below are drawn scattered across the page. Label every node, list each resistor as “node X to node Y”, then find the equivalent resistance and the current through every resistor.

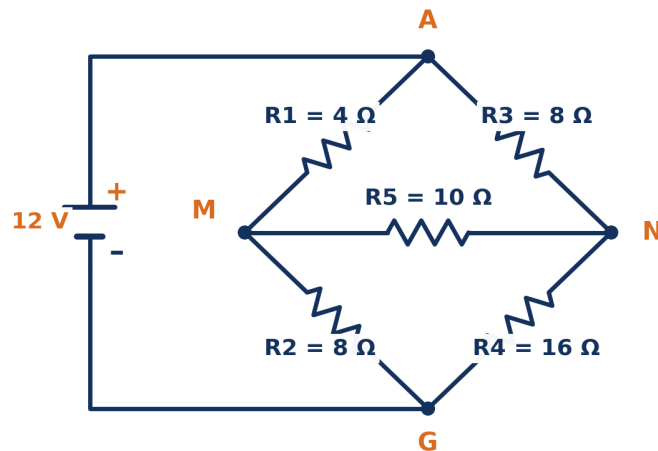


- 19.** A bare wire connects node M to node N.
- (a) Explain in one sentence what that wire does to the four resistors.
 - (b) Find the equivalent resistance and the total current.
 - (c) Find the current through each of the four resistors.

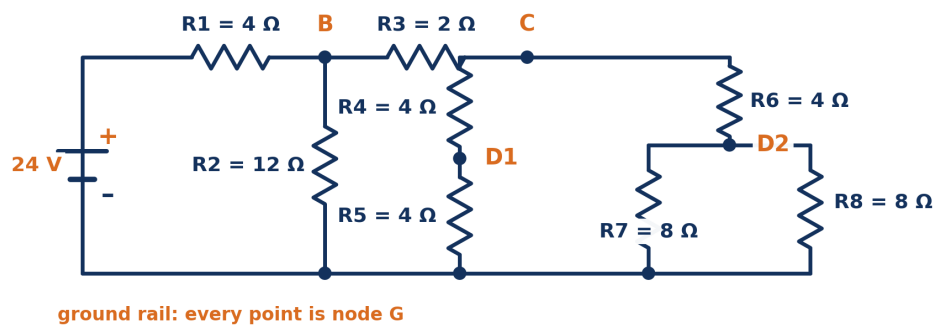


20. R_5 spans node M to node N.

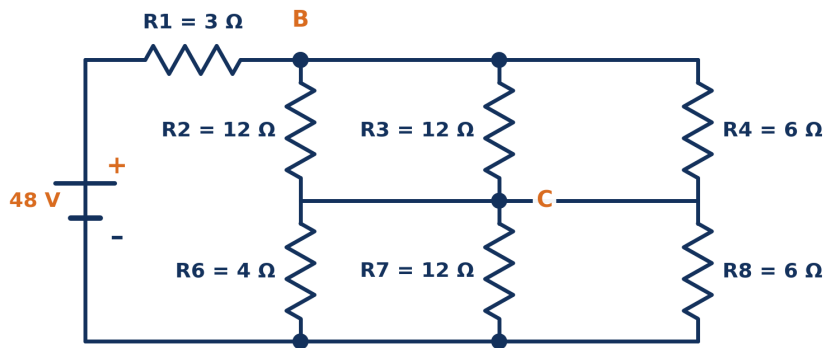
- (a) Find the potential at node M and at node N, measured from node G, using series voltage division only.
- (b) Use your answer to (a) to find the current through R_5 .
- (c) Find the equivalent resistance and the total current from the battery.



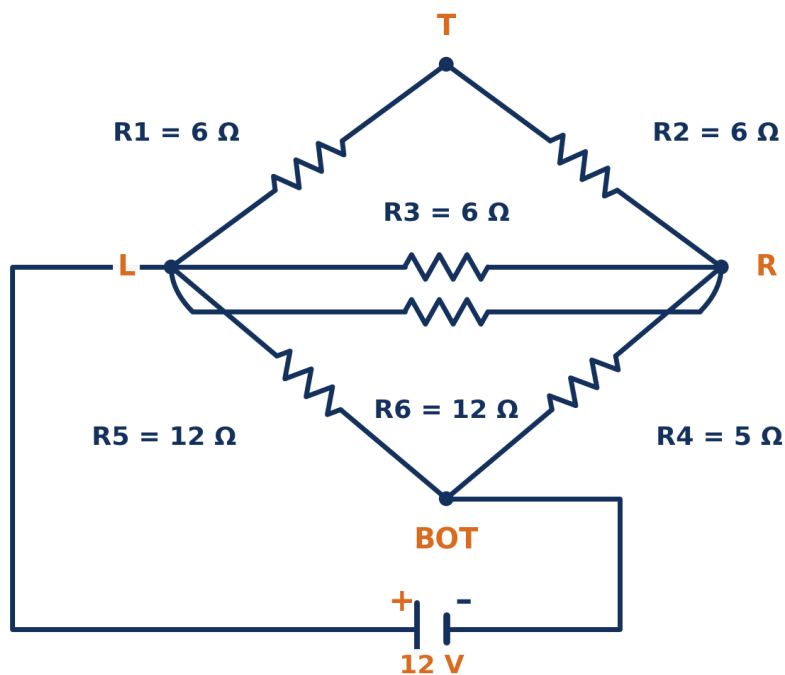
- 21.** Eight resistors. Every point on the bottom rail is the same node G. Collapse the network starting from the far right and work back to the battery. Find the equivalent resistance, the total current, and the current through R_7 .



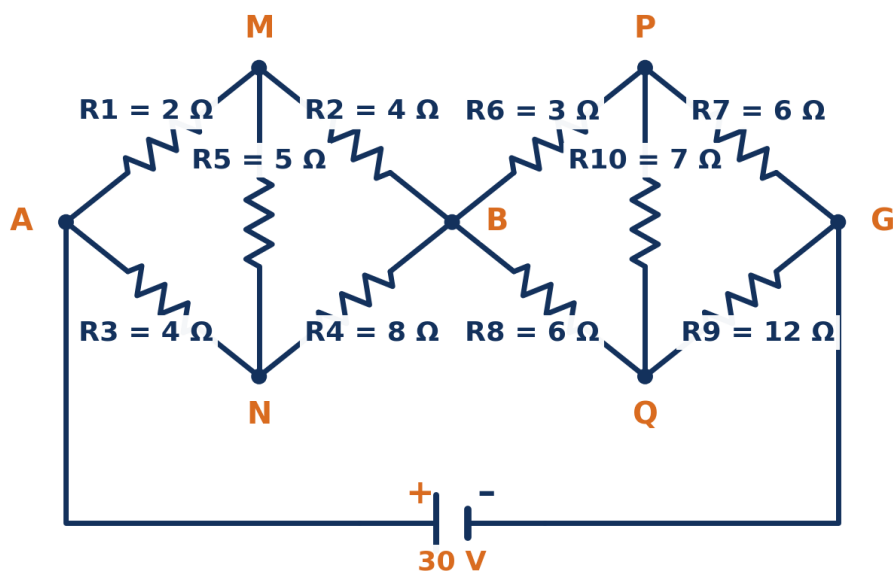
- 22.** This is a complex mixed circuit with multiple parallel branches. Find the equivalent resistance, the total current, and the full IRV chart for all seven resistors.



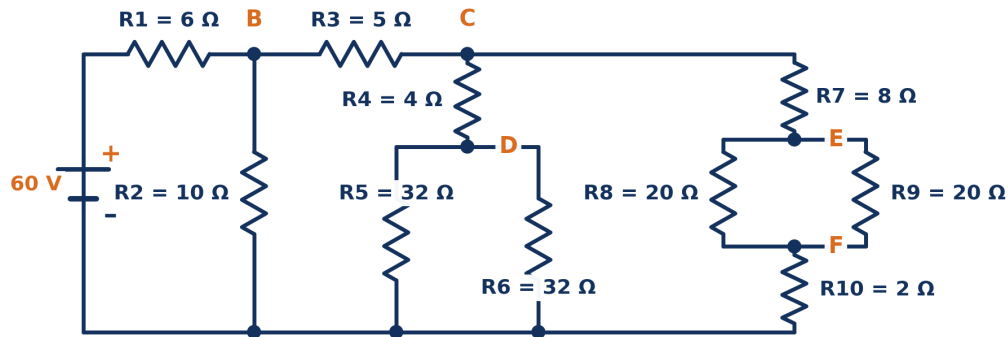
- 23.** The double diamond. R_3 and R_6 are two separate chords, both running from node L to node R.
- Find the equivalent resistance and the total battery current.
 - Find the current and voltage for all six resistors.
 - R_5 burns out and opens. State whether the current through R_4 changes, and justify using nodes.



- 24.** Two bridge sections in series, ten resistors total.
- Show that R_5 and R_{10} each carry zero current, using voltage division on each arm.
 - Find the equivalent resistance and the total current.
 - Find the voltage across R_2 and across R_9 .



- 25.** Ten resistors, multiple parallel branches at three different depths. Find the equivalent resistance, the total current, and the current through R_5 , R_8 , and R_{10} .



- 26.** No diagram. Reasoning only.
- A resistor in a working circuit reads exactly 0 V across it. Give the two physically different reasons this can happen.
 - In a series pair, which resistor dissipates more power, the larger or the smaller? In a parallel pair, which one? Justify both with a formula.
 - Removing one resistor from a mixed circuit changes the current in resistors nowhere near it. Explain why in two sentences.

Answer Key

Standard 7.1 · full worked solutions

1. One resistor, one loop.

(a) $I = V/R = 12/4 = 3.0 \text{ A}$

(b) 12 V. The only other elements are ideal wires, which drop no voltage, so the entire battery voltage must appear across the resistor.

(c) $q = It = (3.0)(5.0) = 15 \text{ C}$

(a) 3.0 A (b) 12 V (c) 15 C

Common error: Thinking some voltage is “lost in the wires”. Ideal wire has zero resistance and therefore zero drop.

2. Series: resistances add and the current is shared.

(a) $R_{\text{eq}} = 3 + 5 = 8.0 \Omega$

(b) $I = 24/8 = 3.0 \text{ A}$, the same through both

(c) $V_1 = (3.0)(3) = 9.0 \text{ V}$; $V_2 = (3.0)(5) = 15.0 \text{ V}$

Check: $9 + 15 = 24 \text{ V}$

$R_{\text{eq}} = 8.0 \Omega$ $I = 3.0 \text{ A}$ $V_1 = 9.0 \text{ V}$ $V_2 = 15.0 \text{ V}$

Common error: Splitting the current between the two resistors. In series the current never divides.

3. Parallel: voltage is shared and currents add.

(a) $R_{\text{eq}} = (6)(9)/(6+9) = 54/15 = 3.6 \Omega$

(b) Both ends of each resistor touch the battery terminals, so each has the full 18 V.

(c) $I_1 = 18/6 = 3.0 \text{ A}$; $I_2 = 18/9 = 2.0 \text{ A}$

$I_{\text{total}} = 3.0 + 2.0 = 5.0 \text{ A}$. Check: $18/3.6 = 5.0 \text{ A}$

$R_{\text{eq}} = 3.6 \Omega$ $V = 18 \text{ V}$ on each $I_1 = 3.0 \text{ A}$ $I_2 = 2.0 \text{ A}$ $I_{\text{total}} = 5.0 \text{ A}$

Common error: Adding 6 + 9 to get 15 Ω . A parallel result must come out below 6 Ω , so anything larger than the smallest branch is instantly wrong.

4. Three in series.

(a) $R_{\text{eq}} = 3 + 6 + 9 = 18.0 \Omega$, $I = 36/18 = 2.0 \text{ A}$

(b) $V_1 = (2)(3) = 6.0 \text{ V}$; $V_2 = (2)(6) = 12.0 \text{ V}$; $V_3 = (2)(9) = 18.0 \text{ V}$. Check: $6 + 12 + 18 = 36 \text{ V}$

(c) R_3 , the largest. All three carry the same current, and $V = IR$, so the largest resistance needs the largest push to drive that shared current through it.

$R_{\text{eq}} = 18.0 \Omega$ $I = 2.0 \text{ A}$ $V = 6.0 / 12.0 / 18.0 \text{ V}$

Common error: Assuming the three resistors split the voltage equally. They split it in proportion to their resistances.

5. Three in parallel.

(a) $1/R = 1/6 + 1/12 + 1/4 = 2/12 + 1/12 + 3/12 = 6/12$, so $R_{eq} = 2.0 \Omega$

(b) Each has the full 36 V:

$I_1 = 36/6 = 6.0 \text{ A}$; $I_2 = 36/12 = 3.0 \text{ A}$; $I_3 = 36/4 = 9.0 \text{ A}$

$I_{total} = 6 + 3 + 9 = 18.0 \text{ A}$. Check: $36/2 = 18 \text{ A}$

(c) R_3 , the smallest. All three have the same voltage across them, and $I = V/R$, so the least opposition carries the most current.

$R_{eq} = 2.0 \Omega$ $I = 6.0 / 3.0 / 9.0 \text{ A}$ $I_{total} = 18.0 \text{ A}$

Common error: Computing $1/R = 6/12$ and stopping there. That is $1/R_{eq}$, not R_{eq} . Invert at the end.

6. Work backwards from the measured current.

(a) $R_{eq} = V/I = 20/2.0 = 10.0 \Omega$

(b) $R_2 = 10 - 4 = 6.0 \Omega$

(c) $V_1 = (2.0)(4) = 8.0 \text{ V}$; $V_2 = (2.0)(6) = 12.0 \text{ V}$. Check: $8 + 12 = 20 \text{ V}$

$R_{eq} = 10.0 \Omega$ $R_2 = 6.0 \Omega$ $V_1 = 8.0 \text{ V}$ $V_2 = 12.0 \text{ V}$

Common error: Dividing the source voltage by R_1 to get a current. Only the equivalent resistance goes with the source voltage.

7. Reasoning and inversions.

(a) $1/4 = 1/12 + 1/x \rightarrow 1/x = 3/12 - 1/12 = 2/12 \rightarrow x = 6.0 \Omega$

(b) $1/R = 3/30 = 1/10$, so $R = 10.0 \Omega$. Identical resistors in parallel: divide by how many there are.

(c) Series components all share the same CURRENT. Parallel components all share the same VOLTAGE.

(d) Adding a branch gives charge a second route, so more total current flows for the same voltage, and more current at the same voltage is exactly what less resistance means.

(a) 6.0Ω (b) 10.0Ω (c) series shares current, parallel shares voltage

Common error: Reversing (c). Getting this backwards makes every later calculation wrong, so check it against the picture: one path means one current.

8. The simplest mixed circuit: collapse the pair, then treat it as series.

(a) $R_2 \parallel R_3 = (4)(4)/8 = 2.0 \Omega$

$R_{eq} = 2 + 2 = 4.0 \Omega$, $I_{total} = 12/4 = 3.0 \text{ A}$

(b) V across $R_1 = (3.0)(2) = 6.0 \text{ V}$

V across the pair = $12 - 6 = 6.0 \text{ V}$

(c) $I_{R2} = 6/4 = 1.5 \text{ A}$; $I_{R3} = 6/4 = 1.5 \text{ A}$. Check: $1.5 + 1.5 = 3.0 \text{ A}$

$R_{eq} = 4.0 \Omega$ $I_{total} = 3.0 \text{ A}$ $V_{R1} = 6.0 \text{ V}$ $V_{pair} = 6.0 \text{ V}$ $I_{R2} = I_{R3} = 1.5 \text{ A}$

Common error: Using the full 12 V to find the branch currents. Only 6 V sits across the pair.

9. R_2 , R_3 and R_4 all span node B to ground. Three in parallel.

$$1/R = 1/6 + 1/12 + 1/4 = 2/12 + 1/12 + 3/12 = 6/12, \text{ so } R = 2.0 \, \Omega$$

$$R_{eq} = 2 + 2 = 4.0 \, \Omega, I_{total} = 36/4 = 9.0 \, A$$

$$V \text{ across } R_1 = (9)(2) = 18.0 \, V, \text{ so } V \text{ across the bank} = 18.0 \, V$$

$$I_{R2} = 18/6 = 3.0 \, A; I_{R3} = 18/12 = 1.5 \, A; I_{R4} = 18/4 = 4.5 \, A$$

$$\text{Check: } 3.0 + 1.5 + 4.5 = 9.0 \, A$$

$$R_{eq} = 4.0 \, \Omega \quad I_{total} = 9.0 \, A \quad I = 3.0 / 1.5 / 4.5 \, A \quad V = 18.0 \, V \text{ on each}$$

Common error: Reading the three angled resistors as a series chain because they are drawn fanning out. All three touch node B and the rail, so they are parallel.

10. Series R_1 feeding a parallel pair.

$$R_2 \parallel R_3 = (3)(6)/9 = 2.0 \, \Omega, R_{eq} = 2 + 2 = 4.0 \, \Omega$$

$$I_{total} = 12/4 = 3.0 \, A$$

$$P_{R1} = I^2 R = (3)^2(2) = 18.0 \, W$$

$$V \text{ across the pair} = 12 - (3)(2) = 6.0 \, V$$

$$P_{R2} = V^2/R = 36/3 = 12.0 \, W; P_{R3} = 36/6 = 6.0 \, W$$

$$\text{Check: } 18 + 12 + 6 = 36 \, W = (3.0)(12)$$

$$P_{R1} = 18.0 \, W > P_{R2} = 12.0 \, W > P_{R3} = 6.0 \, W. R_1 \text{ shares current with the pair; } R_2 \text{ and } R_3 \text{ share voltage.}$$

Common error: Assuming the largest resistor always dissipates the most. In parallel the smaller resistor wins, because it draws more current at the same voltage.

11. Work backwards from the measured total current.

$$R_{eq} = 18/3.0 = 6.0 \, \Omega; \text{ parallel section} = 6 - 2 = 4.0 \, \Omega$$

$$4 = (12)(R_x)/(12 + R_x) \rightarrow 48 + 4R_x = 12R_x$$

$$8R_x = 48 \rightarrow R_x = 6.0 \, \Omega$$

$$V \text{ across } R_1 = (3.0)(2) = 6.0 \, V, \text{ so } 12.0 \, V \text{ sits across the pair}$$

$$I_{R2} = 12/12 = 1.0 \, A; I_{Rx} = 12/6 = 2.0 \, A$$

$$R_x = 6.0 \, \Omega \quad I_{R2} = 1.0 \, A \quad I_{Rx} = 2.0 \, A$$

Common error: Setting R_x equal to the whole $6 \, \Omega$ equivalent without subtracting R_1 first.

12. Start at the branch where the current is known.

$$V \text{ across } R_3 = (1.0)(6) = 6.0 \, V$$

$$R_2 \text{ and } R_3 \text{ are parallel, so } V \text{ across } R_2 = 6.0 \, V \text{ as well}$$

$$I_{R2} = 6/12 = 0.5 \, A; I_{total} = 1.0 + 0.5 = 1.5 \, A$$

$$V \text{ across } R_1 = (1.5)(4) = 6.0 \, V$$

$$V = 6.0 + 6.0 = 12.0 \, V$$

$$V = 12.0 \, V \quad I_{total} = 1.5 \, A \quad V_{R1} = 6.0 \, V$$

Common error: Treating the measured $1.0 \, A$ as the total current. It is one branch of a parallel pair.

13. Opening a switch removes an entire parallel branch.

(a) S open: the R_3 branch is broken, carries no current and offers no path.

$$R_{eq} = 4 + 8 = 12.0 \, \Omega, I = 24/12 = 2.0 \, \text{A}, V_B = (2.0)(8) = 16.0 \, \text{V}$$

(b) S closed: $R_2 \parallel R_3 = (8)(8)/16 = 4.0 \, \Omega$

$$R_{eq} = 4 + 4 = 8.0 \, \Omega, I = 24/8 = 3.0 \, \text{A}, V_B = (3.0)(4) = 12.0 \, \text{V}$$

(c) Adding a resistor in PARALLEL opens a second route for charge. Two routes carry more total current than one, so the equivalent resistance falls and the battery current rises.

Open: 12.0 Ω , 2.0 A, $V_B = 16.0$ V. Closed: 8.0 Ω , 3.0 A, $V_B = 12.0$ V.

Common error: Adding R_3 into the series total when S closes. It is parallel with R_2 , not in series with anything.

14. A branch with a free end is an open circuit.

R_3 has no complete path, so $I_{R3} = 0$, and $V = IR = (0)(10) = 0 \, \text{V}$.

R_3 contributes nothing. $R_2 \parallel R_4 = (8)(8)/16 = 4.0 \, \Omega$

$$R_{eq} = 4 + 4 = 8.0 \, \Omega, I_{total} = 24/8 = 3.0 \, \text{A}$$

$$V_B = 24 - (3.0)(4) = 12.0 \, \text{V}, \text{ so } I_{R2} = I_{R4} = 1.5 \, \text{A}$$

$R_{eq} = 8.0 \, \Omega$ $I_{total} = 3.0 \, \text{A}$ $I_{R3} = 0 \, \text{A}$ $V_{R3} = 0 \, \text{V}$

Common error: Including R_3 in the parallel bank because it is drawn touching the rail. A resistor needs two live connections to matter.

15. A bare wire in parallel with a resistor kills that resistor.

(a) The wire makes B and C the same node. One node has no potential difference with itself, so V across $R_2 = 0$ and $I_{R2} = 0/6 = 0 \, \text{A}$.

(b) $R_3 \parallel R_4 = (6)(12)/18 = 4.0 \, \Omega$

$$R_{eq} = 2 + 0 + 4 = 6.0 \, \Omega, I_{total} = 18/6 = 3.0 \, \text{A}$$

(c) V across $R_1 = 6.0 \, \text{V}$, so 12.0 V across the pair

$$I_{R3} = 12/6 = 2.0 \, \text{A}; I_{R4} = 12/12 = 1.0 \, \text{A}. \text{ Check: } 2 + 1 = 3 \, \text{A}$$

(d) Wire cut: R_2 rejoins in series. $R_{eq} = 2 + 6 + 4 = 12.0 \, \Omega$

$I_{R2} = 0 \, \text{A}$ $R_{eq} = 6.0 \, \Omega$ $I_{total} = 3.0 \, \text{A}$ $I_{R3} = 2.0 \, \text{A}$ $I_{R4} = 1.0 \, \text{A}$ Cut: 12.0 Ω

Common error: Treating the shorting wire as a very small resistance and carrying it forward. An ideal wire is exactly $0 \, \Omega$, so the parallel result is exactly 0.

16. The switch shorts R_2 out of the circuit.

(a) S open: $R_{eq} = 4 + 8 + 12 = 24.0 \, \Omega, I = 24/24 = 1.0 \, \text{A}$

$$V \text{ across } R_2 = (1.0)(8) = 8.0 \, \text{V}$$

(b) S closed: B and C become one node, so V across $R_2 = 0 \, \text{V}$ and $I_{R2} = 0$.

$$R_{eq} = 4 + 12 = 16.0 \, \Omega, I = 24/16 = 1.5 \, \text{A}$$

(c) P open = $(1.0)^2(12) = 12.0 \, \text{W}$; P closed = $(1.5)^2(12) = 27.0 \, \text{W}$

$$\text{Factor} = 27/12 = 2.25, \text{ which is } (1.5/1.0)^2$$

Open: 1.0 A, $V_{R2} = 8.0$ V. Closed: 1.5 A, $V_{R2} = 0$ V. Power in R_3 rises 2.25 $\times$.

Common error: Scaling power with current rather than current squared. $P = I^2R$, so 1.5 $\times$ the current gives 2.25 $\times$ the power.

17. Every spoke runs hub to ring, and the ring is one node.

The ring is bare wire, so all five outer ends are the SAME node. All five resistors span H to that node: five in parallel.

$$1/R = 1/20 + 1/40 + 1/40 + 1/80 + 1/80 = 4/80 + 2/80 + 2/80 + 1/80 + 1/80 = 10/80 = 1/8$$

$$R_{eq} = 8.0 \, \Omega, I_{total} = 24/8 = 3.0 \, A$$

Each resistor sees the full 24 V:

$$I_a = 24/20 = 1.2 \, A; I_b = I_c = 24/40 = 0.6 \, A; I_d = I_e = 24/80 = 0.3 \, A$$

$$\text{Check: } 1.2 + 0.6 + 0.6 + 0.3 + 0.3 = 3.0 \, A$$

$$R_{eq} = 8.0 \, \Omega \quad I_{total} = 3.0 \, A \quad I = 1.2 / 0.6 / 0.6 / 0.3 / 0.3 \, A$$

Common error: Reading the radial layout as a series chain because the spokes are drawn end to end around a circle. They all touch the same two nodes.

18. Node labelling turns a scattered drawing into two parallel banks.

Node list: R_1 : A-B. R_2 : A-B. R_3 : B-G. R_4 : B-G. R_5 : B-G.

$$R_1 \parallel R_2 = (10)(10)/20 = 5.0 \, \Omega$$

$$1/R = 1/10 + 1/20 + 1/20 = 2/20 + 1/20 + 1/20 = 4/20, \text{ so } R = 5.0 \, \Omega$$

$$R_{eq} = 5 + 5 = 10.0 \, \Omega, I_{total} = 20/10 = 2.0 \, A$$

$$V \text{ across each bank} = (2.0)(5) = 10.0 \, V$$

$$I_{R1} = I_{R2} = 1.0 \, A; I_{R3} = 1.0 \, A; I_{R4} = I_{R5} = 0.5 \, A$$

$$R_{eq} = 10.0 \, \Omega \quad I_{total} = 2.0 \, A \quad I_{R1} = I_{R2} = I_{R3} = 1.0 \, A \quad I_{R4} = I_{R5} = 0.5 \, A$$

Common error: Solving straight from the drawing. Write the node list first; the arithmetic becomes trivial once the list exists.

19. The wire merges M and N into a single node.

(a) With M and N merged, R_1 and R_3 both span A to that node, so they are parallel; R_2 and R_4 both span it to G, so they are parallel too.

$$(b) R_1 \parallel R_3 = (6)(12)/18 = 4.0 \, \Omega$$

$$R_2 \parallel R_4 = (6)(12)/18 = 4.0 \, \Omega$$

$$R_{eq} = 4 + 4 = 8.0 \, \Omega, I_{total} = 24/8 = 3.0 \, A$$

$$(c) V \text{ across each pair} = (3.0)(4) = 12.0 \, V$$

$$I_{R1} = 2.0 \, A; I_{R3} = 1.0 \, A; I_{R2} = 2.0 \, A; I_{R4} = 1.0 \, A$$

$$R_{eq} = 8.0 \, \Omega \quad I_{total} = 3.0 \, A \quad I_{R1} = I_{R2} = 2.0 \, A \quad I_{R3} = I_{R4} = 1.0 \, A$$

Common error: Treating this as a bridge needing advanced methods. The connecting element is a wire, not a resistor, so the two midpoints collapse into one node.

20. A balanced bridge: prove the bridge element is dead before touching it.

(a) Ignore R_5 for a moment. Left arm = $4 + 8 = 12 \Omega$, so $V_M = 12 \times 8/12 = 8.0 \text{ V}$.
Right arm = $8 + 16 = 24 \Omega$, so $V_N = 12 \times 16/24 = 8.0 \text{ V}$.

(b) $V_M - V_N = 0$, so $I_{R5} = 0/10 = 0 \text{ A}$. R_5 can be deleted.

This is self-consistent: with no current in R_5 each arm really is a plain divider, which is what we assumed. The ratio test is $R_1/R_2 = 4/8 = 1/2$ and $R_3/R_4 = 8/16 = 1/2$. Equal ratios mean balanced.

(c) $R_{eq} = (12)(24)/36 = 288/36 = 8.0 \Omega$, $I_{total} = 12/8 = 1.5 \text{ A}$ (left arm 1.0 A, right arm 0.5 A)

$$V_M = V_N = 8.0 \text{ V} \quad I_{R5} = 0 \text{ A} \quad R_{eq} = 8.0 \Omega \quad I_{total} = 1.5 \text{ A}$$

Common error: Trying to combine R_5 in series or parallel with something. Until $V_M = V_N$ is proved it is in neither, and once proved, R_5 is simply gone.

21. Collapse the ladder from the far end inward, one layer at a time.

$$R_7 \parallel R_8 = (8)(8)/16 = 4.0 \Omega$$

$$\text{Right branch } C \rightarrow G = R_6 + 4 = 8.0 \Omega$$

$$\text{Left branch } C \rightarrow G = R_4 + R_5 = 8.0 \Omega$$

$$C \rightarrow G \text{ total} = (8)(8)/16 = 4.0 \Omega$$

$$B \rightarrow G \text{ through } R_3 = 2 + 4 = 6.0 \Omega$$

$$B \rightarrow G \text{ total} = (12)(6)/18 = 4.0 \Omega$$

$$R_{eq} = 4 + 4 = 8.0 \Omega, \quad I_{total} = 24/8 = 3.0 \text{ A}$$

$$V \text{ across } R_1 = 12.0 \text{ V, so } V_B = 12.0 \text{ V}$$

$$I_{R2} = 1.0 \text{ A; } I_{R3} = 2.0 \text{ A; } V \text{ across } R_3 = 4.0 \text{ V}$$

$$V_C = 8.0 \text{ V, so each } 8 \Omega \text{ branch carries } 1.0 \text{ A}$$

$$V_{D2} = 8 - (1.0)(4) = 4.0 \text{ V, } I_{R7} = 4/8 = 0.5 \text{ A}$$

$$R_{eq} = 8.0 \Omega \quad I_{total} = 3.0 \text{ A} \quad I_{R7} = 0.5 \text{ A}$$

Common error: Starting the collapse at the battery. A ladder can only be reduced from the end furthest from the source.

22. Two parallel banks in series, seven resistors.

$$\text{Upper bank } B \rightarrow C: 1/R = 1/12 + 1/12 + 1/6 = 4/12, R = 3.0 \Omega$$

$$\text{Lower bank } C \rightarrow G: 1/R = 1/4 + 1/12 + 1/6 = 3/12 + 1/12 + 2/12 = 6/12, R = 2.0 \Omega$$

$$R_{eq} = 3 + 3 + 2 = 8.0 \Omega, \quad I_{total} = 48/8 = 6.0 \text{ A}$$

$$V_{R1} = 18.0 \text{ V; } V \text{ upper bank} = 18.0 \text{ V; } V \text{ lower bank} = 12.0 \text{ V. Check: } 18+18+12 = 48 \text{ V}$$

$$\text{Upper: } I_{R2} = I_{R3} = 1.5 \text{ A, } I_{R4} = 3.0 \text{ A (sum } 6.0 \text{ A)}$$

$$\text{Lower: } I_{R6} = 3.0 \text{ A, } I_{R7} = 1.0 \text{ A, } I_{R8} = 2.0 \text{ A (sum } 6.0 \text{ A)}$$

$$R_{eq} = 8.0 \Omega \quad I_{total} = 6.0 \text{ A} \quad \text{Upper } 1.5 / 1.5 / 3.0 \text{ A at } 18.0 \text{ V} \quad \text{Lower } 3.0 / 1.0 / 2.0 \text{ A at } 12.0 \text{ V}$$

Common error: Combining all six bank resistors into one parallel group. The two banks are separated by node C and are in series with each other.

23. Three separate paths span L to R; R_5 sits straight across the battery.

(a) $R_1 + R_2 = 12\ \Omega$ from L to R (they share node T alone).

That $12\ \Omega$, $R_3 = 6\ \Omega$ and $R_6 = 12\ \Omega$ all span L to R:

$$1/R = 1/12 + 1/6 + 1/12 = 4/12, \text{ so } R = 3.0\ \Omega$$

Chain L→BOT = $3 + 5 = 8.0\ \Omega$; $R_5 = 12\ \Omega$ also spans L to BOT

$$R_{eq} = (12)(8)/20 = 4.8\ \Omega, I_{total} = 12/4.8 = 2.5\ \text{A}$$

(b) R_5 : $12.0\ \text{V}$, $1.0\ \text{A}$. Chain: $12/8 = 1.5\ \text{A}$

R_4 : $V = 7.5\ \text{V}$, $I = 1.5\ \text{A}$; V across the L-R block = $4.5\ \text{V}$

R_3 : $4.5\ \text{V}$, $0.75\ \text{A}$. R_6 : $4.5\ \text{V}$, $0.375\ \text{A}$. R_1 , R_2 : $0.375\ \text{A}$ each at $2.25\ \text{V}$ each

Check: $0.75 + 0.375 + 0.375 = 1.5\ \text{A}$

(c) No change. R_5 connects L to BOT, which are the battery terminals themselves, so its branch is independent. The chain still sees $12\ \text{V}$ and still carries $1.5\ \text{A}$. Only the battery total falls, from $2.5\ \text{A}$ to $1.5\ \text{A}$.

$$R_{eq} = 4.8\ \Omega \quad I_{total} = 2.5\ \text{A} \quad R_1, R_2: 0.375\ \text{A} / 2.25\ \text{V} \quad R_3: 0.75\ \text{A} / 4.5\ \text{V} \quad R_4: 1.5\ \text{A} / 7.5\ \text{V} \quad R_5: 1.0\ \text{A} / 12.0\ \text{V} \quad R_6: 0.375\ \text{A} / 4.5\ \text{V}$$

Common error: Calling R_1 and R_2 parallel because they sit on opposite sides of the diamond. They meet at node T with nothing else attached, which makes them series.

24. Both bridges are balanced, so both bridge resistors vanish.

(a) Left section: $R_1/R_2 = 2/4 = 1/2$ and $R_3/R_4 = 4/8 = 1/2$. Equal, so $V_M = V_N$ and $I_{R5} = 0$.

Right section: $R_6/R_7 = 3/6 = 1/2$ and $R_8/R_9 = 6/12 = 1/2$. Equal, so $V_P = V_Q$ and $I_{R10} = 0$.

(b) Left block = $(6)(12)/18 = 4.0\ \Omega$; right block = $(9)(18)/27 = 6.0\ \Omega$

$$R_{eq} = 4 + 6 = 10.0\ \Omega, I_{total} = 30/10 = 3.0\ \text{A}$$

(c) V across the left block = $12.0\ \text{V}$; upper arm current = $12/6 = 2.0\ \text{A}$, so $V_{R2} = (2.0)(4) = 8.0\ \text{V}$.

V across the right block = $18.0\ \text{V}$; lower arm current = $18/18 = 1.0\ \text{A}$, so $V_{R9} = (1.0)(12) = 12.0\ \text{V}$.

$$I_{R5} = I_{R10} = 0\ \text{A} \quad R_{eq} = 10.0\ \Omega \quad I_{total} = 3.0\ \text{A} \quad V_{R2} = 8.0\ \text{V} \quad V_{R9} = 12.0\ \text{V}$$

Common error: Checking balance by comparing R_1 with R_3 instead of comparing the two RATIOS. Balance needs $R_1/R_2 = R_3/R_4$, not any single pair being equal.

25. Collapse each sub-branch fully before combining anything higher up.

$$R_5 \parallel R_6 = (32)(32)/64 = 16.0 \, \Omega, \text{ so left branch C} \rightarrow \text{G} = 4 + 16 = 20.0 \, \Omega$$

$$R_8 \parallel R_9 = (20)(20)/40 = 10.0 \, \Omega, \text{ so right branch C} \rightarrow \text{G} = 8 + 10 + 2 = 20.0 \, \Omega$$

$$\text{C} \rightarrow \text{G total} = (20)(20)/40 = 10.0 \, \Omega$$

$$\text{B} \rightarrow \text{G through } R_3 = 5 + 10 = 15.0 \, \Omega$$

$$\text{B} \rightarrow \text{G total} = (10)(15)/25 = 6.0 \, \Omega$$

$$R_{\text{eq}} = 6 + 6 = 12.0 \, \Omega, I_{\text{total}} = 60/12 = 5.0 \, \text{A}$$

$$V \text{ across } R_1 = 30.0 \, \text{V}, \text{ so } V_B = 30.0 \, \text{V}$$

$$I_{R2} = 3.0 \, \text{A}; I_{R3} = 2.0 \, \text{A}; V \text{ across } R_3 = 10.0 \, \text{V}$$

$$V_C = 20.0 \, \text{V}, \text{ so each } 20 \, \Omega \text{ branch carries } 1.0 \, \text{A}$$

$$V_D = 20 - 4 = 16.0 \, \text{V}, \text{ so } I_{R5} = 16/32 = 0.5 \, \text{A}$$

$$V_E = 12.0 \, \text{V} \text{ and } V_F = 2.0 \, \text{V}, \text{ so } V \text{ across } R_8 = 10.0 \, \text{V} \text{ and } I_{R8} = 0.5 \, \text{A}$$

$$R_{10} \text{ carries the whole right branch: } 1.0 \, \text{A}$$

$$R_{\text{eq}} = 12.0 \, \Omega \quad I_{\text{total}} = 5.0 \, \text{A} \quad I_{R5} = 0.5 \, \text{A} \quad I_{R8} = 0.5 \, \text{A} \quad I_{R10} = 1.0 \, \text{A}$$

Common error: Combining R_4 with R_7 because both hang off node C. They belong to different branches and never share a second node.

26. Reasoning only.

(a) Reason one: the resistor is shorted, so a bare wire joins its two ends, both ends are the same node, and the difference is exactly zero. Reason two: the resistor carries no current because its branch is open somewhere, and $V = IR = 0$ when $I = 0$.

(b) Series: both carry the same current and $P = I^2 R$, so the LARGER resistance dissipates more.

Parallel: both have the same voltage and $P = V^2/R$, so the SMALLER resistance dissipates more.

(c) Removing a resistor changes the equivalent resistance of its bank, which changes the total current the battery delivers. That new total flows through every series element, so their voltage drops change, which redistributes the voltage across every remaining bank.

(a) shorted, or an open branch (b) series: larger; parallel: smaller (c) the change propagates through the equivalent resistance and the total current

Common error: Answering (b) with one rule for both cases. Which formula applies depends on which quantity is shared, and series and parallel share opposite things.